GROUP 5 | DS & AI LAB PROJECT

SerpentGuard AI: Snake Detection & Classification

Live Video Tracking, Calibrated Species ID and Offline Edge Models

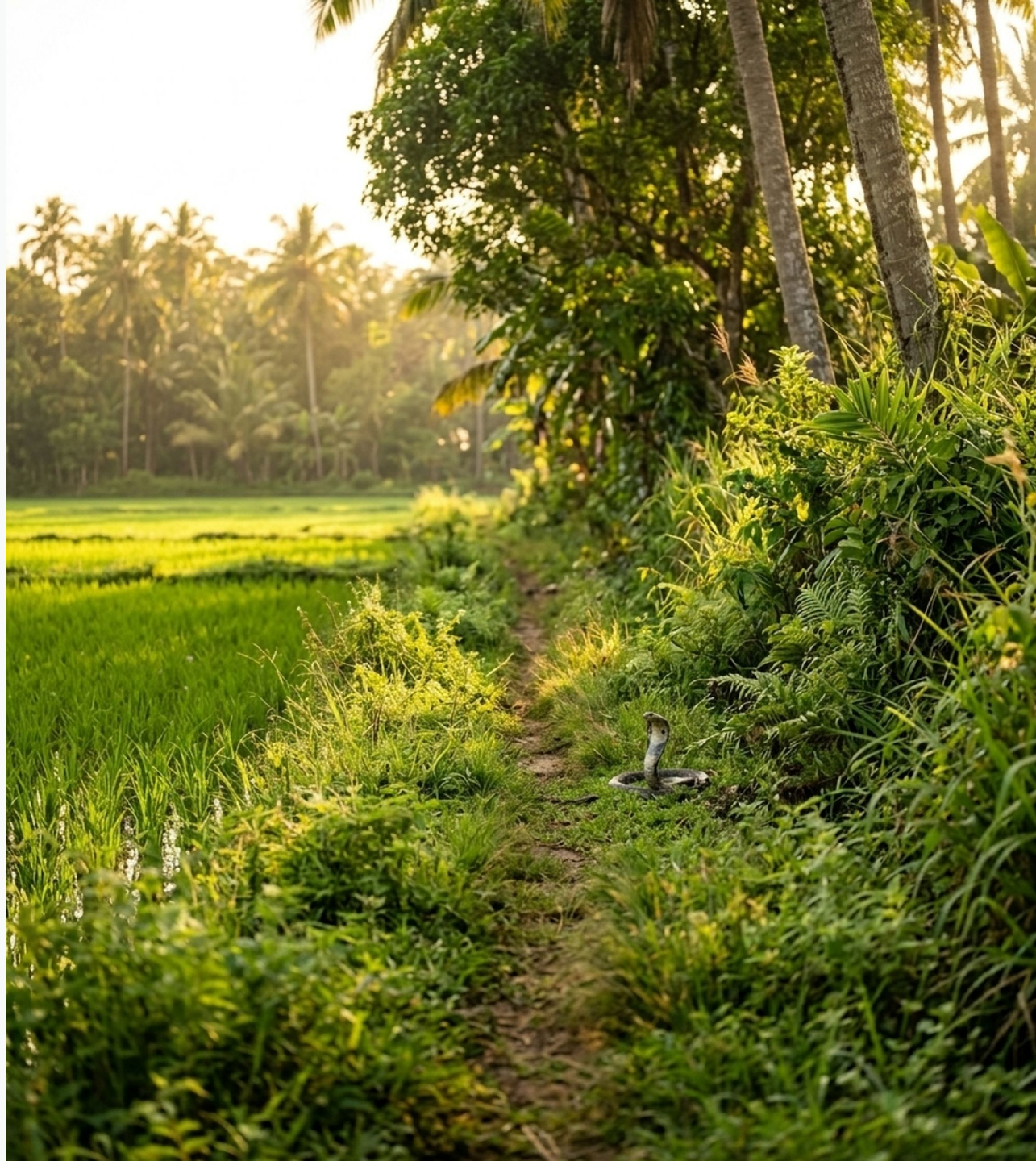
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1. BACKGROUND & MOTIVATION

Why Snakes? Fine-Grained Field AI

Why It Matters

- **Real-World Urgency:** India had an estimated 46,000 snakebite deaths in 2005 (Million Death Study, cited by WHO 2016); encounters trigger panic and snake killings.
- **Fine-Grained Vision:** Look-alike species share patterns; blur, low light, and occlusion break generic models.
- **Safety First:** Encourages a safe distance and never replaces a trip to the hospital.



What We Build, For Whom, and How We Measure It

1. Scope

- **In Scope:** 27 Indian species incl. the Big Four; photos + live video.
- **Out of Scope:** Other species shown as unknown; no bite diagnosis.

2. Stakeholders

- **Users:** Households, farmers, campus staff, snake rescuers.
- **Also:** Forest dept., clinicians (education only), researchers.

3. Targets (M5)

- **Accuracy:** Macro-F1 ≥ 0.80; top-3 ≥ 0.93; detector P and R ≥ 0.90.
- **Safety & Speed:** Dangerous-species recall ≥ 0.95; ECE ≤ 0.05; ≥ 10 FPS.

3. COMPARISON WITH EXISTING ALTERNATIVES

How SerpentGuard AI Differs From Cloud Search & Live LLMs

Capability	Google Search / Lens	Gemini / ChatGPT Live	Direct Cloud LLMs	SerpentGuard AI (Ours)
Primary Input	Static image upload	Cloud video streaming	Static prompt & image	**Real-time local camera feed**
Rural Offline Use	Needs internet	Needs a live connection	Needs internet + API	**Offline edge model (target)**
Fine-Grained Focus	General web matches	Generalist visual prior	Can sound sure when wrong	**27 Indian species, fixed list**
Safety Abstention	Returns visual matches	No measured abstention	No calibrated confidence	**Calibrated abstain + safety KB**

4. ACADEMIC LITERATURE & IMPLEMENTATION GAP

Beyond SnakeCLEF & YOLO Studies

What We Do Differently From Papers

- **SnakeCLEF (2021-2024):** Strong ViT/CNN models, evaluated on static photos; live on-device video is outside the benchmark.
- **Prior YOLO Studies (2024-25):** 87-88% detection precision; we found no calibrated abstention in them.
- **Two-Stage Edge Design:** Couples real-time snake detection with calibrated fine-grained classification, offline.



5. SYSTEM ARCHITECTURE & OFFLINE DEPLOYMENT

Live Video, Frame by Frame, Fully Offline

1. Edge Detector

- **YOLOv8n (int8):** Finds snakes in every live video frame.
- **ByteTrack:** Keeps one ID per snake across frames.

2. Classifier

- **MobileViT:** 27 Indian species; full fine-tune vs LoRA.
- **Abstention Gate:** Calibrated thresholds + voting over recent frames.

3. Safety KB

- **No LLM APIs:** No cloud latency or cost; no generated text.
- **Local Lookup:** Species facts + official first-aid text, verbatim.

6. APPLICATION UI MOCKUPS (REAL-TIME EDGE EXPERIENCE)

SerpentGuard AI Interface: Live Detection & Abstention UI

Screen A: High-Confidence Detection

- **Live ROI & Top-3 Match:** Illustrative: 1. Spectacled Cobra (0.86) | 2. Rat Snake (0.09) | 3. Russell's Viper (0.03).
- **Visual & Habitat Cues:** Hood mark & dorsal pattern; facts from offline species KB.
- **Safety Quick Actions:** [Official First-Aid Steps] and [Call Local Rescuer].

Screen B: Uncertain / Abstain Alert

- **Abstention Triggered:** Confidence below threshold due to motion blur or foliage occlusion.
- **Look-Alike Caution:** Ambiguous match between Common Krait and non-venomous Wolf Snake.
- **Mandatory Safety Guard:** Do not approach snake: [Keep Safe Distance] and [Emergency Guide].



7. INVESTIGATING MODEL LIMITATIONS & EDGE CASES

Testing Occlusion & Calibration

Evaluation Protocol

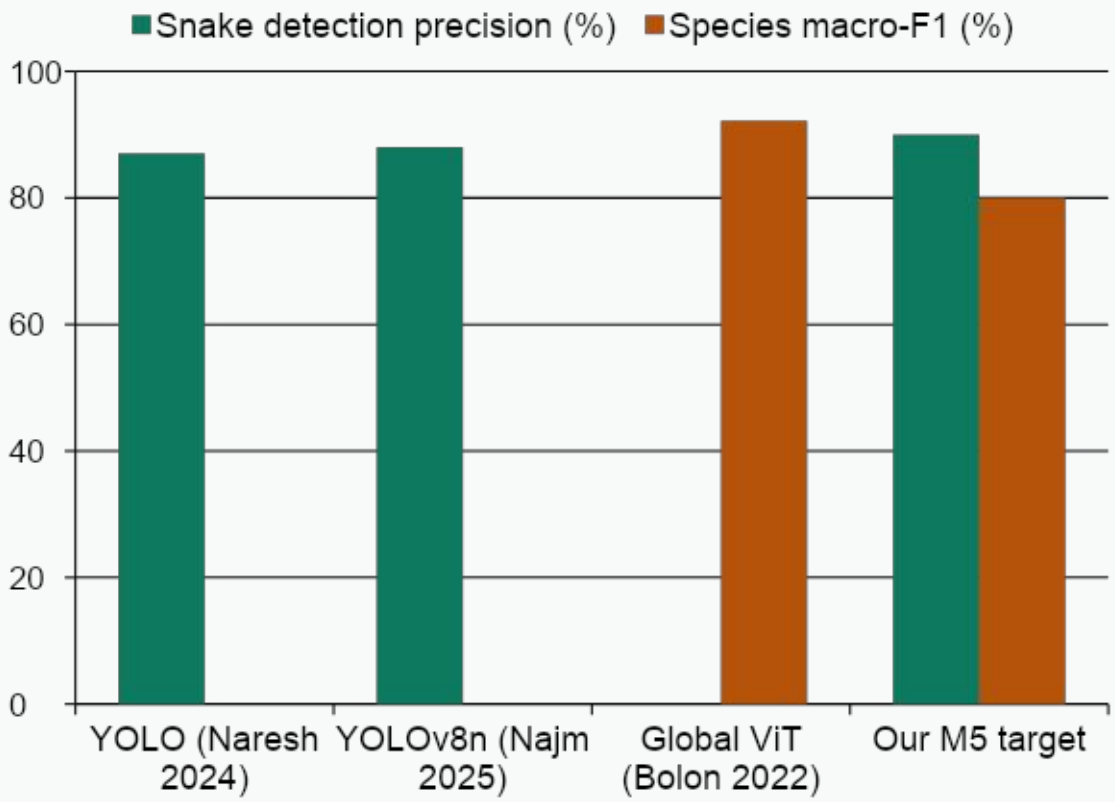
- **Look-Alike Confusion:** Quantifies confusion matrices between venomous snakes (Krait, Cobra) and mimics.
- **Degraded Visibility:** Measures recall drops under motion blur, night light, and synthetic 30-70% occlusion.
- **Calibration Metrics:** Evaluates macro-F1, per-species recall, and Expected Calibration Error (ECE).

8. ROADMAP & EVALUATION TARGETS

Build the Vision Pipeline First

Research Contributions

- **Core Vision First:** The live frame-by-frame pipeline runs end-to-end on a data subset at Milestone 3.
- **Ablations:** Benchmarks LoRA fine-tuning and two-stage cropping against single-stage baselines.
- **Safety Checks:** Tests abstention so a missed detection is never shown as a safe area.



Team Review and Acknowledgment

By signing, each member confirms they have reviewed this presentation, understand it, and agree with the submission.
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